Retail Sales Performance Analysis & Business Insights

Analytics Report & Insights Documentation | Internship Project

Dataset: Sample Superstore | Analysis period: 2014–2017 | Prepared: October 2026

1. Executive Summary

This report evaluates 9,994 Superstore transaction lines to identify commercial performance drivers, margin pressure, and actionable improvement opportunities. The dataset records $2,297,200.86 in sales and $286,397.02 in profit, equivalent to a 12.47% profit margin. Technology leads category profit ($145,454.95), the West region leads total profit ($108,418.45), and Tables are the largest loss-making subcategory (−$17,725.48). These findings favor targeted margin improvement, careful discount governance, and closer review of weak-performing product lines. The dataset is historical and does not establish current performance or causal effects.

2. Business Objectives & Stakeholders

Business objective: evaluate sales, profitability, product mix, regional differences, customer segments and discount patterns to inform retail pricing and merchandising decisions. Intended stakeholders include sales leadership, finance, merchandising, regional managers, and marketing teams. Key questions: Which categories and regions contribute most to sales and profit? Where are losses concentrated? How do discounts correlate with margin? Which segments merit attention?

3. Data & Methodology

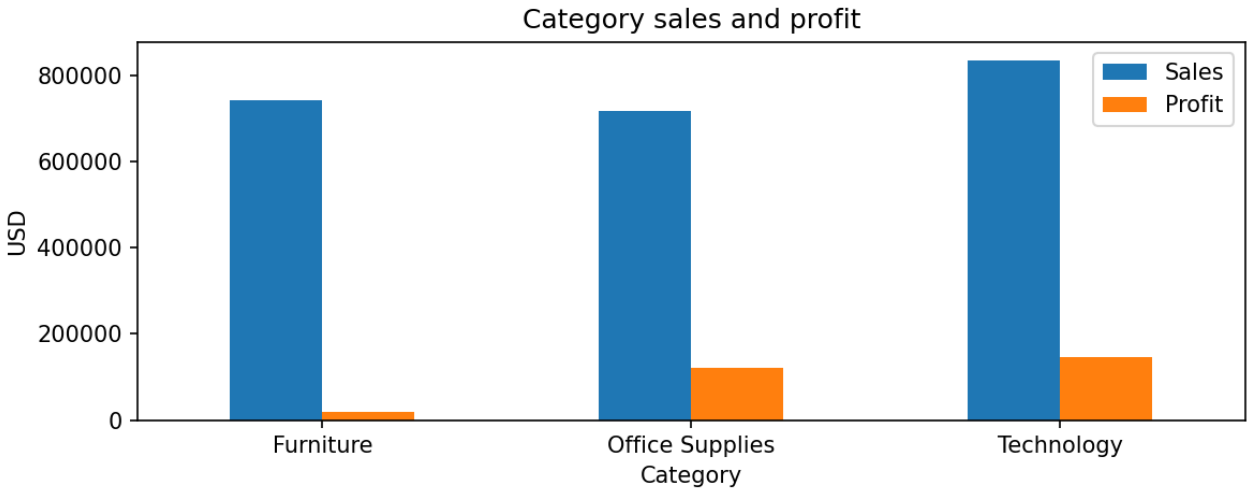
Source: user-provided Sample - Superstore.csv. The 9,994-row, 21-column transactional dataset includes order identifiers, dates, products, regions, customer segments, sales, quantity, discounts and profit. Python with Pandas, NumPy, Matplotlib and Seaborn supports cleaning, descriptive aggregation and charting. Order dates are parsed as dates; missingness, duplicates and types should be reviewed. Each row is an order line, so total orders are counted by unique Order ID rather than row count. Profit margin = total profit / total sales × 100; average order value = total sales / unique orders. This is descriptive, observational analysis, not an experiment.

4. Key Performance Indicators

KPI	Value
Sales	$2,297,200.86
Profit	$286,397.02
Profit margin	12.47%
Unique orders	5,009
Unique customers	793
Units sold	37,873
Average order value	$458.61

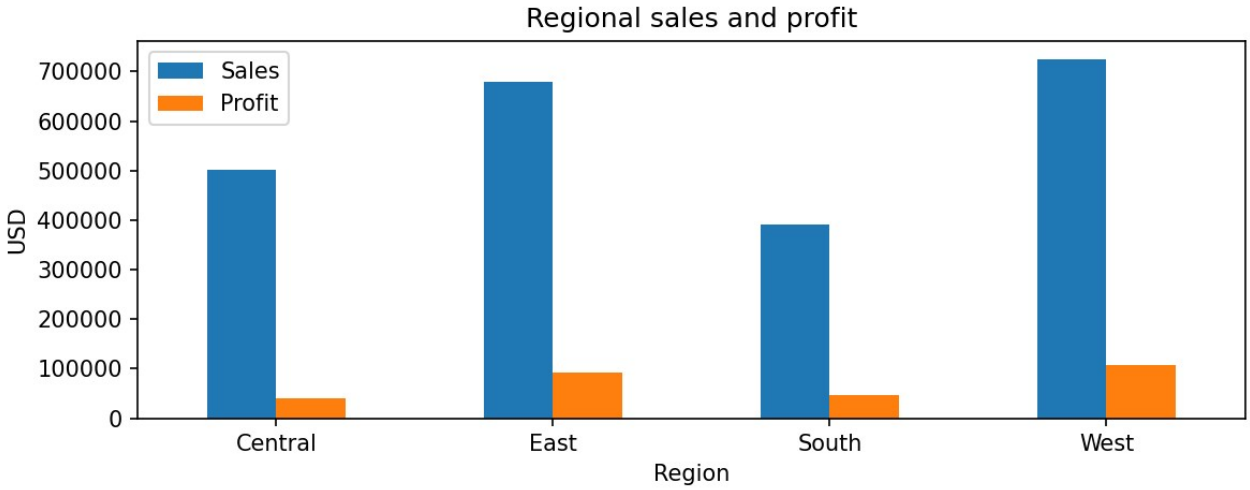
5. Findings & Visual Evidence

5.1 Category profitability



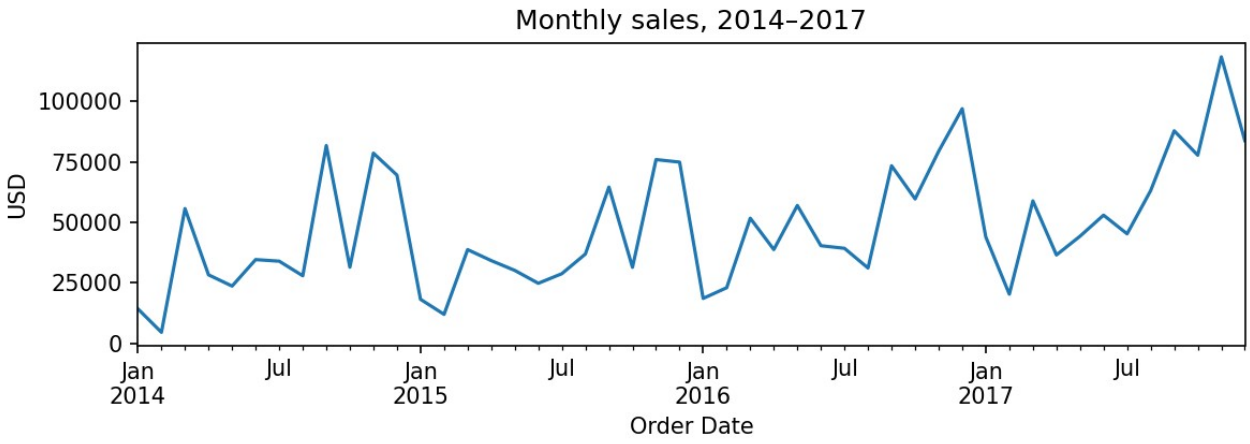
Technology contributes the largest category profit, while Furniture has a relatively weak profit contribution. Category revenue alone is therefore insufficient to judge performance.

5.2 Regional performance



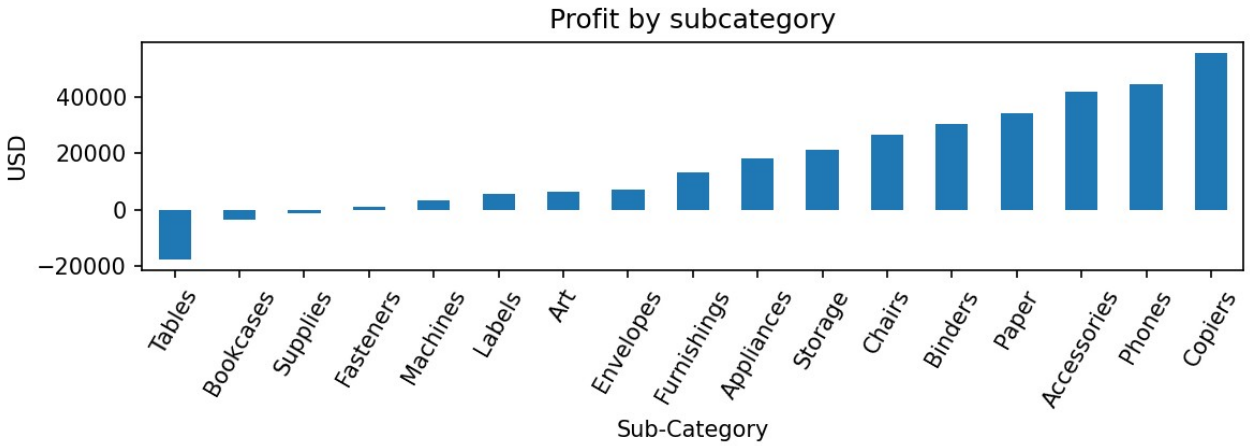
The West region leads sales ($725,457.82) and profit ($108,418.45). Central has the lowest profit margin (7.92%), warranting review of its product mix and discounting.

5.3 Sales trend



Monthly revenue fluctuates substantially across the observed four years. Seasonal interpretation should be supported by month-of-year comparisons and not assumed from one visual alone.

5.4 Subcategory risks



Tables (−$17,725.48), Bookcases (−$3,472.56), and Supplies (−$1,189.10) generate negative aggregate profit. Product-level costs, pricing, shipping and promotions should be investigated before any discontinuation decision.

5.5 Customer and discount analysis

Customer segment	Sales (USD)	Profit (USD)
Consumer	1,161,401.34	134,119.21
Corporate	706,146.37	91,979.13
Home Office	429,653.15	60,298.68

Consumer contributes the largest absolute sales volume, but the profitability of each segment should also be assessed as a margin, not only in dollars. Discount band analysis below is correlational:

Discount band	Sales (USD)	Profit (USD)	Margin
None	1,087,908	320,988	29.5%
1–20%	846,522	100,785	11.9%
>20–50%	298,541	-58,817	-19.7%

>50%	64,229	-76,559	-119.2%

Discount bands mix different products and regions, so negative margins in higher-discount groups do not prove discounts caused the losses.

6. Business Insights & Implications

• Profit is unevenly distributed: Technology is a major contributor, whereas Furniture and selected subcategories depress overall returns.

• Geography matters: West performs strongly; Central needs a more granular review of product mix, discount rates, and transaction economics.

• Customer concentration: Consumer contributes the largest revenue pool, creating a meaningful opportunity for retention and cross-sell analysis.

• Promotions require discipline: high-discount transactions warrant product- and region-adjusted margin review rather than blanket discount elimination.

7. Prioritized Recommendations

Priority	Action	Owner / measure
High	Audit Tables, Bookcases and Supplies by SKU, discount and freight cost	Merchandising / net profit per SKU
High	Test discount guardrails for loss-making combinations	Pricing / margin by discount band
High	Diagnose Central region performance against comparable product mix	Regional sales / margin gap
Medium	Pilot targeted retention and cross-sell for profitable customer cohorts	Marketing / repeat purchase and margin
Medium	Review Technology assortment and stock allocation	Merchandising / contribution margin
Medium	Refresh a monthly sales, margin and loss dashboard	Finance / KPI trend

8. Implementation & Evaluation

Days 1–30: validate the data, investigate negative-profit SKUs and define discount thresholds. Days 31–60: pilot pricing or merchandising changes in a limited set of products or regions, with a comparison group where feasible. Days 61–90: evaluate changes in net profit, profit margin, sales volume, and customer outcomes; expand only where evidence supports benefit. Monitor monthly thereafter.

9. Limitations & Conclusion

This dataset covers 2014–2017, contains historical transaction lines rather than complete customer lifecycles, and lacks detailed unit costs, returns and marketing spend. Consequently, proposed

actions are hypotheses to test, not guaranteed outcomes. The strongest near-term opportunities are reviewing loss-making subcategories, understanding Central region margins, and evaluating discounts using contribution profit alongside revenue.

Appendix: Deliverables

Notebook: Retail_Sales_Analysis.ipynb; analytical tables: Business_KPI_Summary.csv, Category_Analysis.csv, Region_Analysis.csv, Segment_Analysis.csv, Subcategory_Analysis.csv, Discount_Analysis.csv, Business_Insights.csv; workbook: Retail_Business_Analytics.xlsx; eight charts in Visualizations/. These notebook outputs are maintained separately from this report.